THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS V — MATHEMATICS · MODEL PAPER 05 ANSWER KEY

Answers and marking points | Maximum Marks: 80

UNDERSTANDING SHAPE AND SPACE — Marks 21

Q.	Marks	Answer / marking points
1	1	Each small square is 1 cm × 1 cm. The area of the shaded figure is **(b) 14 sq cm**
2	6	Angles A, B and C are shown below. **(i)** (a) Which angle is acute? (b) Which angle is the greatest? (… [3] (a) **A** (b) **C** (c) 90° − 55° = **35°** **(ii)** Three cubes of side 1 cm are placed in a row to form a cuboi… [3] (a) **6** (b) **12** (c) 3 × 6 = 18, minus 4 hidden squares (2 joints × 2) = **14**
3	6	**(i)** Name the solid: (a) it has no vertex and no edge, only one c… [3] (a) sphere (b) cylinder (c) cuboid **(ii)** Which solid does each object look like? (a) a brick (b) a ro… [3] (a) cuboid (b) cylinder (c) sphere
4	8	The plan of a village library is shown. Study it and answer. **(a)** Find the area of the Reading Room. [2] 7 × 5 = **35 sq m** **(b)** What is the total area of the Store and the Computer room? [2] 9 + 12 = **21 sq m** **(c)** Which two rooms have the same area? What is it? [2] **Store and Court**, each 3 × 3 = **9 sq m** **(d)** A carpet costing ₹ 30 per sq m is laid in the Reading Room. … [2] 35 × 30 = **₹ 1,050**

NUMBER SYSTEM — Marks 9

Q.	Marks	Answer / marking points
5	1	800000 + 40000 + 600 + 3 = **(c) 8,40,603**
6	1	56,498 rounded off to the nearest thousand is **(d) 56,000**
7	1	The smallest 6-digit number having all different digits is **(b) 1,02,345**
8	6	**(i)** Sunita ate 2/9 of a cake and her sister Tara ate 4/9 of it. … [3] (a) 2/9 + 4/9 = 6/9 = **$\frac{2}{3}$** (b) 1 − 2/3 = **$\frac{1}{3}$** **(ii)** Write the fraction or mixed number shown by A, B and C: A = … [3] A = $\frac{1}{3}$, B = $1\frac{2}{3}$, C = $2\frac{1}{3}$

OPERATIONS — Marks 25

Q.	Marks	Answer / marking points
9	1	1000 ÷ 25 = ? **(c) 40**
10	3	Raju bought a cricket bat for ₹ 650 and 3 balls at ₹ 85 each. The shopkeeper gave him a di… Balls = 3 × 85 = ₹ 255; Bill = 650 + 255 = ₹ 905 Pays = 905 − 100 = **₹ 805**
11	3	Divide 2,547 by 6. Write the quotient and the remainder and check your answer. Quotient = **424**, Remainder = **3** Check: 6 × 424 + 3 = 2544 + 3 = 2547 ✓

Q.	Marks	Answer / marking points
12	6	**(i)** For Diwali, a family bought 4 kg sweets at ₹ 320 per kg, cra… [3] Sweets = ₹ 1,280; Diyas = 12 × 45 = ₹ 540; Total = 1,280 + 850 + 540 = ₹ 2,670 Left = 3,000 − 2,670 = **₹ 330** **(ii)** (a) Estimate 7,486 + 5,318 by rounding each to the nearest t… [3] (a) 7,000 + 5,000 = **12,000** (b) **12,804**
13	6	**(i)** Bulbs 1 to 28 are connected to switches A and B. Glowing bul… [3] (a) **4** (b) **7** (c) **28** (LCM of 7 and 4) **(ii)** (a) Write all factors of 28. (b) Write the common factors of… [3] (a) 1, 2, 4, 7, 14, 28 (b) 1, 2, 7, 14 (c) **14**
14	6	**(i)** In this number pyramid every number is the sum of the two nu… [3] (a) **8 and 7** (b) **30** (c) 3 + 5 + 2 + 6 = **16**; 30 − 16 = **14** **(ii)** A verandah floor is covered with square tiles of side 1 m as… [3] (a) 10 × 3 = **30** (b) 2 × (10 + 3) = **26 m** (c) 30 × 25 = **₹ 750**

MEASUREMENT — Marks 17

Q.	Marks	Answer / marking points
15	3	Fill in the blanks: (a) 8 km 50 m = _____ m (b) 3.5 kg = _____ g (c) 2 days = _____ hou… (a) 8050 (b) 3500 (c) 48
16	3	Write True or False: (a) 22:30 hours is 10:30 p.m. (b) A 500 g packet is heavier than a 1 … (a) True (b) False (500 g is half a kg) (c) True
17	3	A train leaves Jaipur at 6:50 a.m. and is to reach Delhi at 11:15 a.m. (a) How long is the… (a) 6:50 a.m. to 11:15 a.m. = **4 h 25 min** (b) 11:15 + 1 h 35 min = **12:50 p.m.**
18	8	The map shows some roads of northern Rajasthan with distances (not to scale). **(a)** How far is Jhunjhunu from Jaipur if one goes through Sikar? [2] 114 + 55 = **169 km** **(b)** Which city is nearest to Sikar? How far is it? [2] **Jhunjhunu**, 55 km (Churu is 70 km, Jaipur 114 km) **(c)** A bus goes from Jaipur to Churu through Sikar and comes back… [2] (114 + 70) × 2 = **368 km** **(d)** How much longer is the route Sikar → Jhunjhunu → Churu than … [2] 55 + 80 = 135 km; 135 − 70 = **65 km**

DATA HANDLING AND PATTERNS — Marks 8

Q.	Marks	Answer / marking points
19	1	Which of these is a triangular number (1, 3, 6, 10, 15, …)? **(a) 6**
20	1	Squares are made with matchsticks (Fig. 1 needs 4 sticks, Fig. 2 needs 7, Fig. 3 needs 10)… **(b) 31**
21	3	Students of Class 5 voted for their favourite game. The tally table is shown (a group of f… **(a)** Write the number of votes for Cricket, Kabaddi, Football and… [1] Cricket 14, Kabaddi 9, Football 6, Kho-kho 11 **(b)** Which game got the fewest votes? Which got the most? [1] Fewest: **Football** (6); most: **Cricket** (14) **(c)** How many students voted in all? [1] 14 + 9 + 6 + 11 = **40**
22	3	Look at the pattern: 1, 2, 4, 7, 11, 16, ___, ___, … (a) What is added each time? (b) Writ… (a) The number added goes up by 1 each time: +1, +2, +3, +4, +5, … (b) **22, 29** (c) 29 + 8 = **37**

END OF ANSWER KEY — Total: 80 Marks